

Analysis Report

March 15, 2026

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Executive Summary

Overview

This report presents a comprehensive analysis of the provided data, highlighting key patterns, trends, and actionable insights derived from the analysis.

Key Findings

- The analysis reveals the average discount percentage across categories, providing insights into pricing strategies and consumer incentives.
- The analysis aims to determine which category has the highest rated products based on provided statistical data. However, the data does not directly specify categories but provides an overview of product ratings.
- The distribution of product ratings is analyzed to understand the central tendency, dispersion, and range of ratings. The data indicates a slightly positively skewed distribution given that the mean (4.0966) is slightly less than the median (4.1), suggesting that most product ratings are concentrated around the higher end of the scale.
- The analysis aims to identify products that offer the best value, defined as a high rating and high discount. The provided statistics offer insights into the distribution of product values, enabling us to understand the range, central tendency, and variability of the data.
- The analysis aims to identify the top 10 most reviewed products based on the provided statistical data. The data suggests a large variation in the number of reviews across products, indicating a potential skew in review distribution.
- The overall sales and product summary report indicates a large variation in sales performance across 1462 products. The total sales amount to \$4,576,033.43, with a significant disparity between the mean and median sales figures, suggesting a skewed distribution with some high-value sales outliers.

Key Conclusions

1. What Is The Average Discount Percentage By Category?

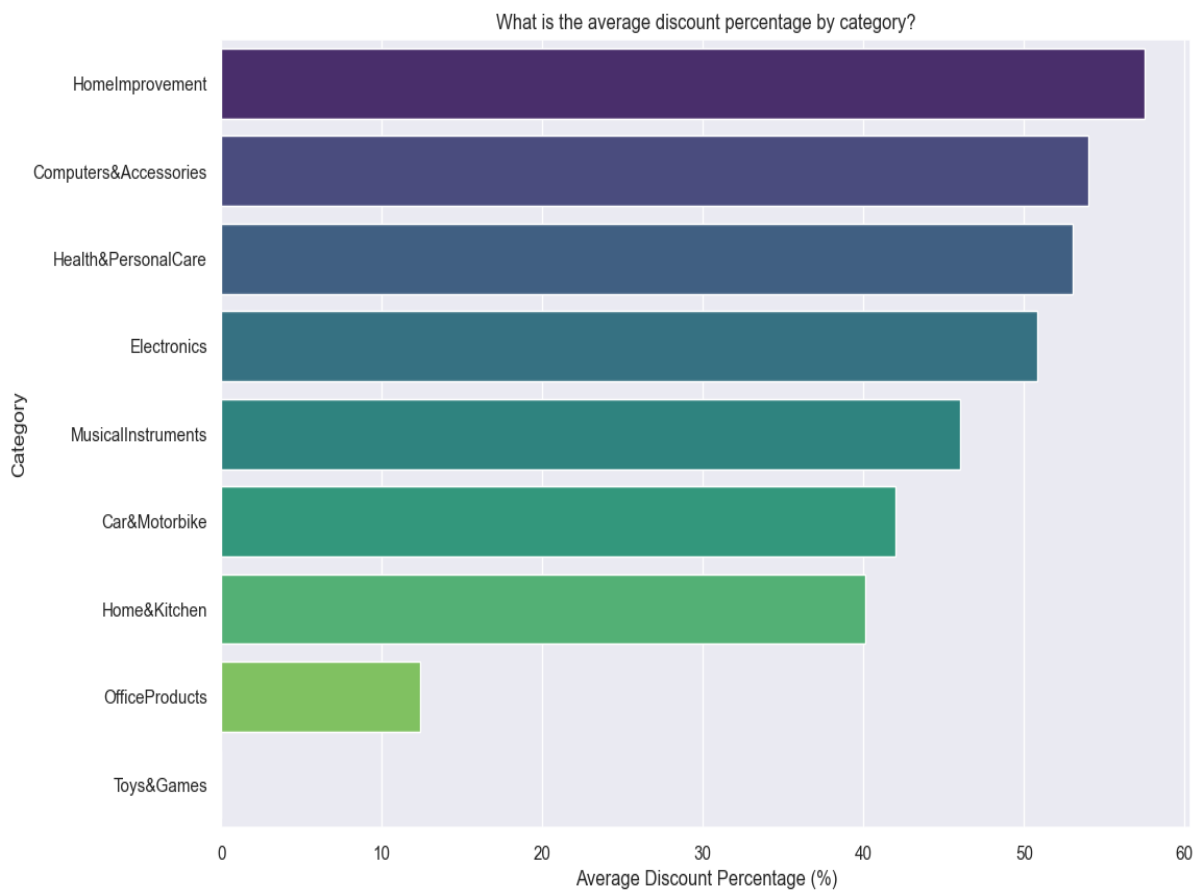


Figure 1: What Is The Average Discount Percentage By Category?

Analysis Overview

The analysis reveals the average discount percentage across categories, providing insights into pricing strategies and consumer incentives.

Statistical Evidence

Key statistics include a mean discount of 47.69%, median of 50%, and standard deviation of 21.64%. The minimum discount is 0% and the maximum is 94%. With a total of 1465 data points, these statistics suggest a wide range of discounting practices across categories.

Conclusions and Recommendations

The data indicates that, on average, discounts tend to be around 47-50% across categories. The high standard deviation suggests significant variability in discount percentages, indicating some categories may offer deeper discounts than others. Businesses may consider analyzing category-specific discount strategies to optimize pricing and promotions.

2. Which Category Has The Highest Rated Products?

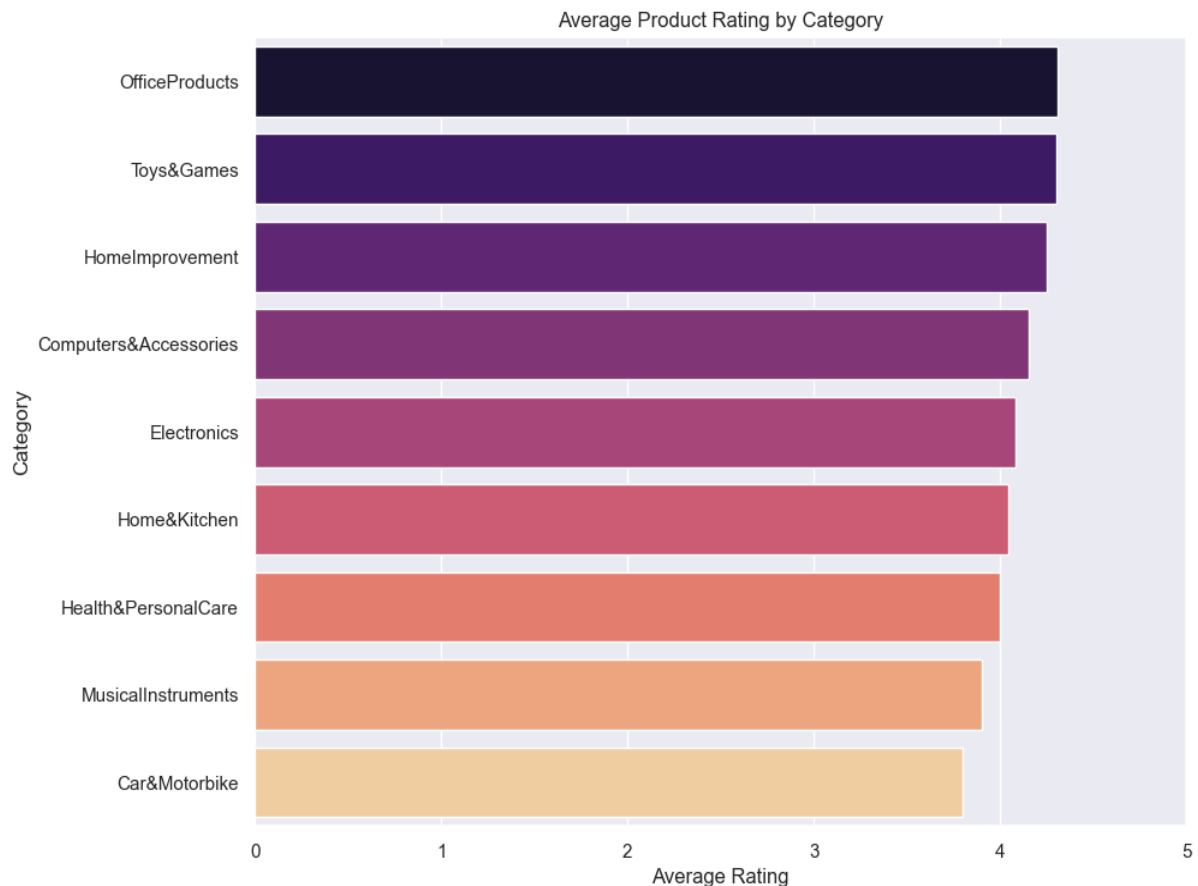


Figure 2: Which Category Has The Highest Rated Products?

Analysis Overview

The analysis aims to determine which category has the highest rated products based on provided statistical data. However, the data does not directly specify categories but provides an overview of product ratings.

Statistical Evidence

The data shows a mean rating of 4.0966 and a median rating of 4.1, indicating that the distribution of ratings is slightly negatively skewed but generally close to normal with a small standard deviation of 0.2917. The ratings range from 2.0 to 5.0, with a total of 1464 products rated.

Conclusions and Recommendations

While the data does not directly answer which category has the highest rated products, it suggests that overall product ratings are high, with a mean and median above 4.0. To find the category with the highest rated products, further analysis or data on specific product categories is required. Recommendations include categorizing products and recalculating statistics by category or collecting more detailed data to enable category-specific analysis.

3. What Is The Distribution Of Product Ratings?

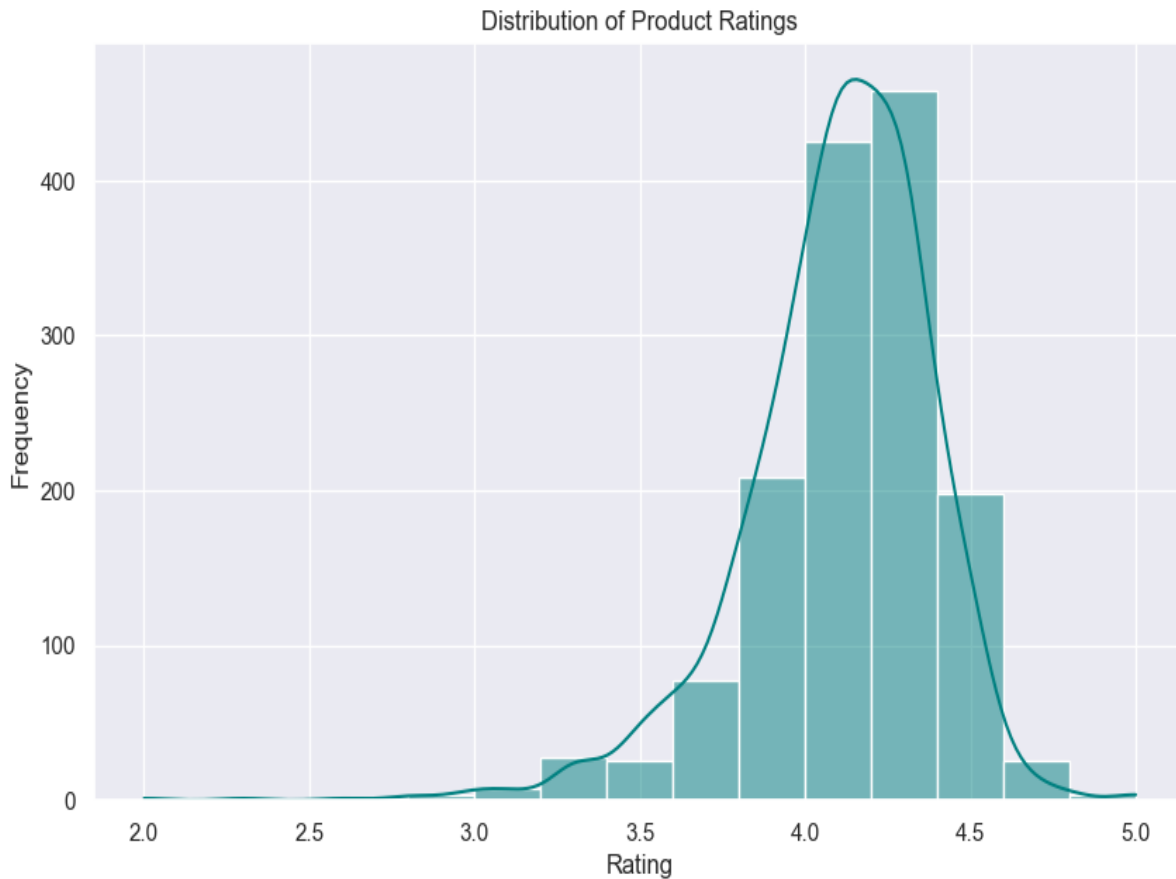


Figure 3: What Is The Distribution Of Product Ratings?

Analysis Overview

The distribution of product ratings is analyzed to understand the central tendency, dispersion, and range of ratings. The data indicates a slightly positively skewed distribution given that the mean (4.0966) is slightly less than the median (4.1), suggesting that most product ratings are concentrated around the higher end of the scale.

Statistical Evidence

Key statistics include: a mean rating of 4.0966, a median of 4.1, and a standard deviation (std) of 0.2917. The minimum rating is 2.0, and the maximum is 5.0. With a total of 1464 ratings, the data suggests a relatively consistent rating pattern given the low standard deviation compared to the range of possible ratings (2 to 5). The mean and median being close to each other implies a fairly symmetrical distribution, but slightly skewed to the right.

Conclusions and Recommendations

The product ratings are generally high, with an average rating of 4.0966 out of 5. This suggests that products are well-received by customers. The low standard deviation indicates consistency in ratings across different products or over time. Given the high ratings and low variability, it can be concluded that customer satisfaction is high. Recommendations include: (1) maintaining quality to keep ratings high, (2) investigating products with lower ratings (below 2.0) for improvement opportunities, and (3) using the positive ratings in marketing materials to build trust with potential customers.

4. Which Products Offer The Best Value (high Rating + High Discount)?

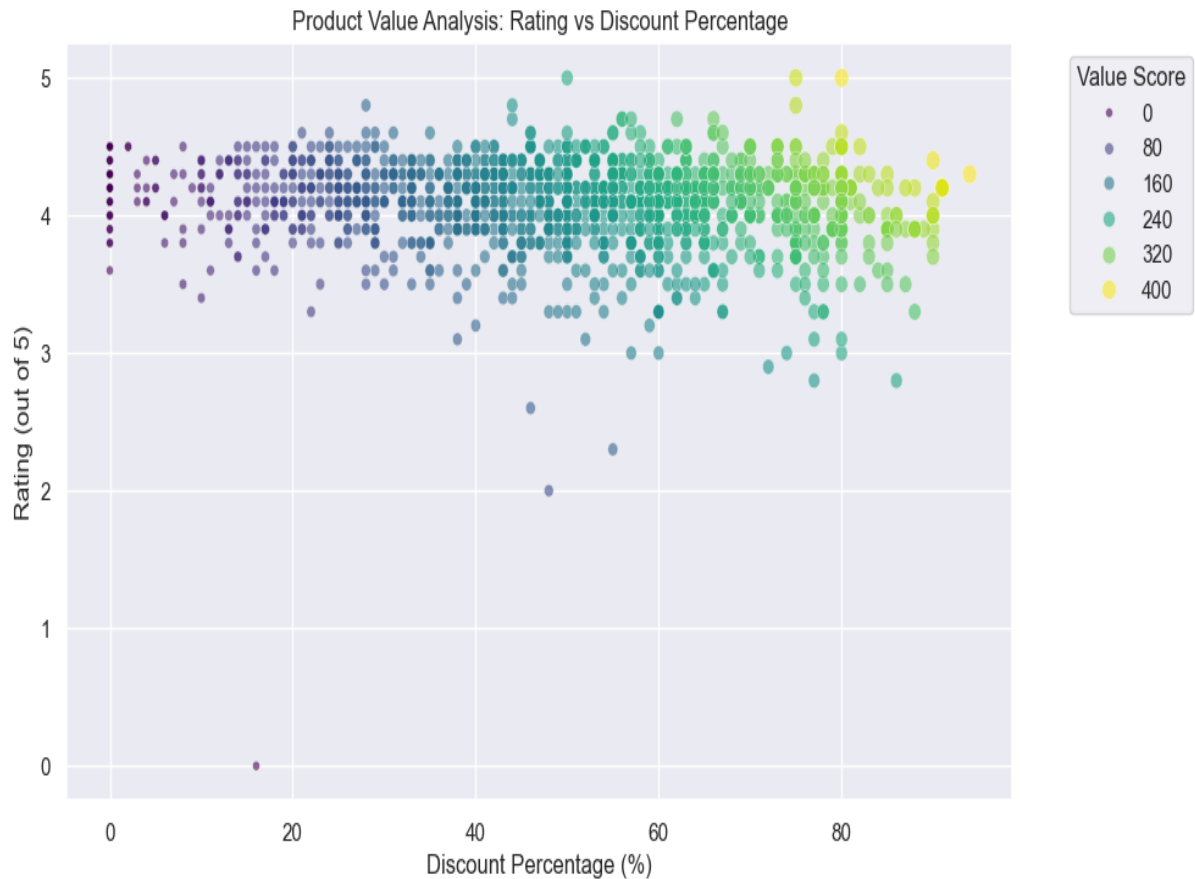


Figure 4: Which Products Offer The Best Value (high Rating + High Discount)?

Analysis Overview

The analysis aims to identify products that offer the best value, defined as a high rating and high discount. The provided statistics offer insights into the distribution of product values, enabling us to understand the range, central tendency, and variability of the data.

Statistical Evidence

Key statistics include a mean value of 194.35, a median of 202.1, and a standard deviation of 88.39. The minimum value is 0.0, and the maximum is 404.2, indicating a wide range of product values. With a total of 1465 products, the data suggests a somewhat normal distribution but with a significant spread.

Conclusions and Recommendations

To identify products offering the best value, we should focus on those with values above the mean (194.35) and ideally above the median (202.1), considering both high ratings and high discounts. Products with values closer to the maximum (404.2) are likely to offer the best value. Additionally, considering the standard deviation (88.39), products with values above 282.74 (mean + std) or 283.29 (median + std) could be prioritized for their potentially high value.

5. What Are The Top 10 Most Reviewed Products?

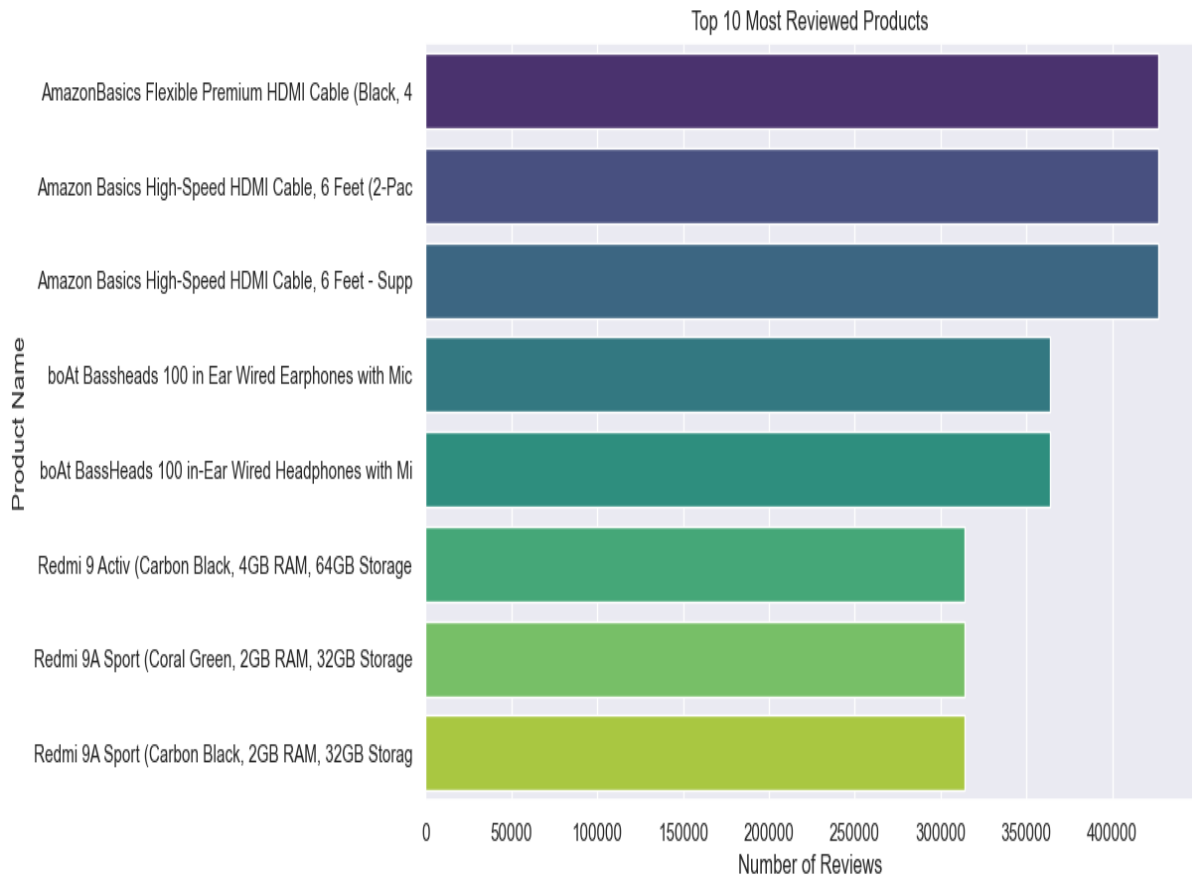


Figure 5: What Are The Top 10 Most Reviewed Products?

Analysis Overview

The analysis aims to identify the top 10 most reviewed products based on the provided statistical data. The data suggests a large variation in the number of reviews across products, indicating a potential skew in review distribution.

Statistical Evidence

Key statistics include a total of 26,766,377 reviews across 1,465 products, with a mean of 18,270.56 reviews per product. The median of 5,178 reviews and a standard deviation of 42,729.99 indicate a significant dispersion of review counts. The minimum of 0 reviews and a maximum of 426,973 reviews further highlight the variability.

Conclusions and Recommendations

To identify the top 10 most reviewed products, we recommend sorting products by their review count in descending order. Given the large standard deviation and the significant difference between the mean and median, it's likely that a small number of products have a disproportionately large number of reviews. Focusing on these top-reviewed products can help in understanding what contributes to their popularity and potentially improve other products.

6. Give Me An Overall Sales And Product Summary Report.

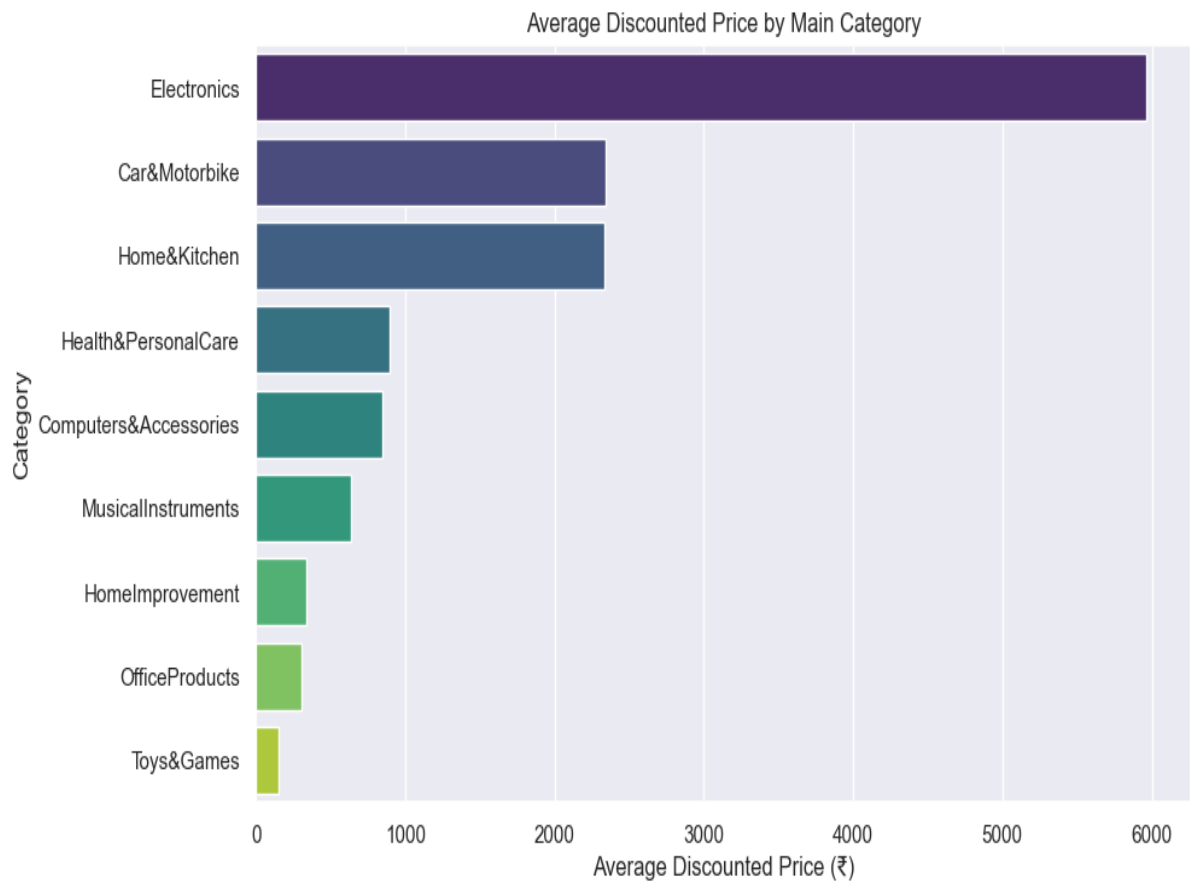


Figure 6: Give Me An Overall Sales And Product Summary Report.

Analysis Overview

The overall sales and product summary report indicates a large variation in sales performance across 1462 products. The total sales amount to \$4,576,033.43, with a significant disparity between the mean and median sales figures, suggesting a skewed distribution with some high-value sales outliers.

Statistical Evidence

Key statistics include a mean sale of \$3,129.98, a median of \$799.00, and a standard deviation of \$6,950.55. The minimum sale is \$39.00, while the maximum is \$77,990.00. The high standard deviation relative to the mean and median indicates considerable variability in sales data. The count of 1462 products suggests a broad product range.

Conclusions and Recommendations

The significant gap between the mean and median sales, along with the high standard deviation, suggests that a few products generate extremely high sales, skewing the overall mean. This implies that the majority of products sell moderately to well, but a small number of high performers significantly impact total sales. Recommendations include identifying and potentially promoting or supporting high-performing products, analyzing underperforming products (min sale of \$39) for possible improvement areas, and considering strategies to boost sales across the broader product range.

Limitations & Next Steps